SIADS 696 Milestone II Project Report

Flight Delay Prediction

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Introduction

Flight delays and cancellations remain a major challenge in air travel, creating uncertainty for passengers and operational difficulties for airlines and airports. These disruptions can lead to missed connections, increased costs, and significant inconvenience for travelers. In this project, we addressed this problem by developing a machine learning system that predicted whether a flight would be delayed, estimated the length of the delay in minutes, and predicted the likelihood of cancellation. We also explored broader patterns in the data by grouping airports, airlines, and routes according to shared delay behavior and weather sensitivity.

Addressing this problem has important practical value for both travelers and the aviation industry. More accurate delay and cancellation predictions can help passengers make better decisions about scheduling, trip planning, and whether flight insurance is worth purchasing. At the same time, identifying larger patterns in disruptions can provide useful operational insight into which airports or routes are most vulnerable to congestion, weather, or other recurring factors. These findings can support better planning, resource allocation, and risk awareness.

We were m otivated to work on this problem because flight delays are common, personally relevant, and influenced by many interacting variables, including weather, seasonality, air traffic, and airline operations. This made the topic both a meaningful real world problem and a strong machine learning challenge. We were also interested in going beyond a standard prediction task by combining supervised and unsupervised learning to better understand the hidden structure of delay and cancellation patterns.

For the supervised portion of the project, we used flight data from the US Bureau of Transportation Statistics together with NOAA weather data to train models of Logistic Regression, XGBoost, and Neural Networks. These models were used to classify whether a flight would be delayed, estimate delay length, and predict cancellation risk. For the unsupervised portion, we applied K-Means and Hierarchical Clustering to identify groups of time of day, airport, airline, and route with similar operational and weather-related patterns. Compared with related projects that focused on a single airport or only used supervised learning, our project contributed a wider regional analysis and incorporated unsupervised methods to reveal hidden relationships among airports and routes, improving both interpretability and overall insight.

Related Work

We discovered a similar project done that develops an ML model utilizing similar datasets for flight metadata from Kaggle.com while also cross-referencing with Climate.gov to obtain weather data using neural networks, random forests, Gradient Boosted trees, and K-Nearest Neighbors for O'Hare International airport during the year of 2015.

While we plan to look at similar metrics, our project differs in that we are attempting to gather data from airports in different regions, while the other project focused primarily on gathering data from one region during different seasons. Furthermore, they utilized only forms of supervised learning using ML modeling, while we will attempt to boost our model by adding an unsupervised learning aspect to identify more hidden patterns among airports, airlines, and flight routes.

Another project that provided valuable insight into our project is from Enhancing Flight Delay Predictions Using Network Centrality Measures, which provides a similar clustering model approach by using network centrality measures such as K-Means to cluster airports based on time delay patterns before crafting the prediction model. While we have a similar concept, our project focuses on more aspects of flight data in a more concentrated area(Chicago), looking at metrics such as airline, weather, time of day, and routes.

One other source we reviewed that provided similar examples is a project here Principal Component Analysis & Clustering with Airport Delay Data. This project provides insight into using PCA and clustering on airport operation metrics focusing on taxiing in and out of the airport for flights. It provided an excellent source of information on PCA and how best to structure the data to best visualize the results, however the project was more focused on specific airport efficiency quotes based on time to gate and taxiing, while our project focuses more on weather related aspects.

By reviewing similar projects of work in relation to flight delay, we noticed that most of them focus primarily on supervised machine learning models that predict whether a flight will be delayed using historical flight, airline, airport, and weather data, while ours differs since we combine these results with unsupervised clustering to identify hidden patterns. Rather than simply predicting delay outcomes, we also group airports, airlines, routes, time-of-day, and weather information based on historical delay behavior. This truly allows us to comprehend broader patterns such as which routes are more affected by weather, and which airline-time combinations show similar risk.

Data Sources

We used three primary data sources for this project, the U.S. Bureau of Transportation Statistics (BTS) airline data and historical weather data access through the Meteostat python library, which provides weather records derived from NOAA based sources, and airport metadata downloaded as a CSV file from the U.S. Department of Transportation. The BTS dataset served as the main source for flight operational data, while the weather dataset was used to add environmental conditions that may influence delays and cancellations, and the airport metadata file provided contextual airport information such as geographic coordinates and regional identifiers used during feature engineering and weather matching.

The BTS airline dataset was obtained as a structured tabular dataset in CSV format and contained individual flight records. Important variables from this dataset included scheduled departure and arrival times, airline identifiers, origin and destination airports, and distance-related variables. For our project, we focused on flights associated with O'Hare International Airport (ORD) during January 2026. Before cleaning and preprocessing, the flight dataset contained 26,119 records.

The weather data was accessed programmatically in python using the Meteostat library, which returns historical weather observations in a dataframe compatible format that can be easily merged with pandas. Through this source, we obtained airport related weather variables such as temperature, precipitation, dew point, snow, wind speed, humidity, pressure, and general weather condition codes. These variables were matched to the flight data so that weather conditions at the relevant airport and time could be incorporated into both the supervised and unsupervised analyses.

The airport metadata dataset contained airport level reference information. Important variables from this file included airport identifiers, city, state code, FAA region code, site type, and geographic coordinates. These fields were used to merge contextual information for both origin and destination airports and, most importantly, to retrieve hourly weather observations for the correct airport locations.

Together, these datasets allowed us to combine operational flight information with historical weather conditions and airport level geographic metadata for a specific airport and time period. The final merged dataset represented January 2026 flights at ORD, with each record containing both flight characteristics, matched weather-related predictors, and relevant airport metadata related to delay analysis.

Feature Engineering

The preprocessing workflow began by standardizing the raw flight data so it could be used consistently across both the supervised and unsupervised analyses. Column names were cleaned, blank entries were converted to missing values, and important date and numeric variables were cast into usable formats. The dataset was then filtered to include only flights departing from O'Hare International Airport (ORD), and cancelled, diverted, or incomplete records missing essential fields such as the departure delay label, scheduled departure time, origin, destination, or flight date were removed. These steps reduced noise and ensured that the remaining observations were suitable for modeling.

Several time-based features were then engineered from the scheduled departure and arrival times. Scheduled departure time was split into hour and minute components, converted to minutes after midnight, and encoded cyclically using sine and cosine transformations. Broader departure time bins were also created, along with a weekend indicator, week-of-year and day-of-year variables, and a route feature formed by combining the origin and destination airport codes. Additional airport metadata was merged into the dataset so that origin and destination records included contextual airport information needed for later processing.

Weather variables were then aligned to each flight using the scheduled departure hour. Separate origin and destination weather features were added for temperature, dew point, humidity, precipitation, snow, wind speed, pressure, and weather condition code. This produced a combined record for each flight containing both flight characteristics and environment conditions at the relevant time.

Care was taken to prevent data leakage in the supervised learning stage. Variables that would only be known after departure or arrival were removed, including actual departure and arrival times, delay minutes, taxi times, air time, and post flight delay cause variables. The final supervised feature set therefore focused only on information that would realistically be available before departure, such as scheduled time variables, route and calendar information, distance measures, airport context, and weather conditions.

The unsupervised analysis used the same cleaned base dataset but included several additional derived features designed to capture broader operational patterns. These included bad weather indicators, morning and late-flight flags, time of day categories, and a weather sensitivity measure based on the difference between delay rates in bad weather and normal weather. The data was then aggregated across several levels, including airport, airline, route and time period, so that clustering methods could identify higher level behavioral groupings rather than individual flight patterns.

Supervised Learning

For the supervised learning portion of the project, various models were used to evaluate the ability to predict if flights were going to be delayed or not. Following the cleaning of the data, a logistic regression model, XGBoost classification model, and multilayer perceptron neural network models were implemented to identify the optimal classification method for determining if a flight would experience delays.

The DepDel15 variable in the data is a binary variable indicating if a flight experienced a delay of 15 or more minutes. Initial exploratory model implementation utilized 13 predictor variables. After running the baseline logistic regression model, it became apparent that several of the 13 variables played little to no role in predicting if a flight was delayed or not based on their coefficients in the model summary being very close to zero. This initial exploration led to the 8 variables used as predictors across all three models: CRSElapsedTime, DistanceGroup, sched_dep_sin, sched_dep_cos, origin_temp, origin_prcp, origin_dwpt, and origin_snow. In the same model evaluation, it was determined that origin temperature, origin precipitation, and origin dew point had the greatest impact on predicting if flights were delayed, having lasso regression-smoothed coefficients of 0.2464, 0.2011, and 0.1506 respectively. The other five variables were included to increase model complexity to ensure that the models were not being overfit on just the three most significant variables. In the final optimal model, origin_snow was dropped due to a coefficient of 0.

The first model explored was the logistic regression model. This model was chosen due to its classification styles. A logistic regression model, once trained, predicts the probability of a binary outcome being either positive or negative. The model offers flexibility on the threshold for an outcome, which was appealing for this project, given that a majority of flights do not experience delays, the data is imbalanced. The speed and efficient optimization involved with logistic regression made this model the first one chosen to conduct exploratory analysis and identify a "baseline" for comparing the other two models.

The second model explored was the XGBoost Classification model. The XGBoost classification model is a culmination of decision trees. A decision tree is useful in this project when trying to classify a binary variable. XGBoost utilizes hundreds of decision trees and taking the determined threshold's percentage of decision trees that classify the variable one way compared to the other. The XGBoost classification model is useful because of its efficiency and processing power, its ability to handle missing data, its ability to provide more insight into feature importance, and its ability to handle large amounts of data with ease.

The third and final model explored was a multilayer perceptron (MLP) neural network. The MLP was included in this project because of its ability to learn data complexities that most linear models, like logistic regression, can miss. By utilizing the ReLU activation function, MLPs can create multidimensional relationships amongst data through the various layers that are included. Unlike XGBoost, MLPs allow every feature to interact with each other because of the complex layering involved. Like logistic regression, the MLP outputs the probability of the predictors being one of the two binary outcomes.

For the data handled in the supervised learning section, a StandardScaler was utilized to scale the data accordingly. Since the values of different variables have stark contrast in values (for example, temperature and precipitation), the StandardScaler helps create a more normalized data set, where all variables are scaled to have a mean of zero and standard deviation of one. This adjustment is used to enhance all model performances.

Hyperparameter tuning was conducted on each of the three models in an effort to conduct the best fit algorithm. For the logistic regression model, the lasso regression L1 penalty and liblinear solver was applied to help normalize the features used in the model. The C parameter that measures regularization strength was evaluated to find the value that produced the highest F1 score. This was included because it adds a secondary regularization technique to the logistic regression model to better handle the relationship between variables of different numerical ranges. The most important feature included in the tuning of the logistic regression model was setting the class_weight parameter to 'balanced.' Due to the extreme difference in the quantity of flights that did not experience delays (roughly 70%) compared to those that did at the O'Hare airport in January, the models all struggled with their recall scores due to the imbalance of the training data. The balanced class weight feature makes the penalty greater on the model for those incorrect classifications on flights that did experience a delay, improving the recall of the model. The max iterations of the logistic regression model were tested by looping through values from 100 to 1000 in increments of 50. The iterations of 1000 produced the highest recall, as it allowed the model to run longer without stopping.

In the XGBoost model, the scale_pos_weight parameter was included to add a penalty to the model – similar to the balanced class_weight parameter in the logistic regression model – for when the model incorrectly classifies the minority group (delayed flights). Scale_pos_weight is the fraction of negative cases (non-delayed) divided by the positive cases, adding more emphasis on the negative cases. The logloss eval_metric was added to assess how close the predicted probability of a classification is to its actual classification. The logloss function penalizes vastly incorrect classifications more. The max_depth parameter of 4 is included to increase the number of splits that the decision trees can make en route to arriving at the classification. When the depth is too small, underfitting occurs, and when the depth is too large, overfitting can occur, so a depth of 4 produces a strong medium for fitting. The n_estimators parameter is included to determine the number of decision trees to include in the estimation, and the majority of decision trees' classification becomes the model's classification for a given flight. The subsample is included to introduce random samples of the training data to each decision tree to produce more variability in the classification and prevent overfitting.

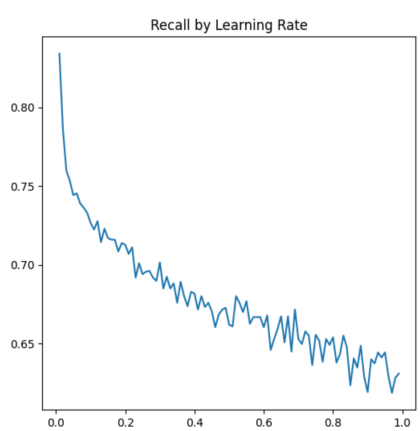
In the MLP neural network model, the num_inputs is set to 8 to match the number of predictors used in the model. The MLP is constructed with an input layer, then a hidden layer with 64 neurons and a ReLU activation function, then another hidden layer with 32 neurons and a ReLU activation function, before passing the content into the final binary classification layer. The three-layer model of the neural network utilizes 64 neurons to start before compressing to 32 and then to 1 to prevent drastic overfitting of the data. 64 serves as an efficient option with (8) times expansion on the number of input features. The Adam optimizer is utilized to dynamically update weights throughout the training process. The learning rate is included similar to how it is used in XGBoost, being optimized to determine the brevity of the training. The pos_weight is added to better handle the ratio of flights that did not experience delays compared to those that did experience delays. The loss function of BCEWithLogitsLoss is used to create a binary classification model with the ensuing entropy that is seen in neural networks, similar to the logloss function used in the other models. Lastly, in the model evaluation phase, epochs are used with a time limit to improve model efficiency. Once a model with strong enough metrics for use is identified, the grid search can stop so that memory and runtime is not overused.

The initial evaluation of the models began with accuracy as the baseline metric. After reviewing classification reports, however, it became clear that due to the imbalanced data, the model would predict that a flight was not delayed, making the recalls incredibly low. The recall of initial logistic regressions was 0.09. Due to this discovery, recall, precision, and the F1 score were determined to be the best metrics to evaluate the supervised models and their performance. As an F1 score is the mean between precision and recall, this will serve as the primary metric to evaluate the model's performance. In the table below, the precision, recall, and F1 scores for the respective models are included as a result of 5-fold cross validation evaluation. The accuracy is included as well to illustrate that it is not the best metric to judge these models on.

Metric	Logistic Regression	XGBoost	MLP
Accuracy	0.6083 ± 0.0073	0.6502 ± 0.0050	0.6419 ± 0.0028
Recall	0.6466 ± 0.0117	0.6951 ± 0.0099	0.8902 ± 0.1042
Precision	0.3993 ± 0.0072	0.4415 ± 0.0043	0.3482 ± 0.0038
F1 Score	0.4937 ± 0.0083	0.5400 ± 0.0036	0.5038 ± 0.0064

Following the model parameter analysis and feature importance exploration, the models produced the results seen above. The XGBoost model performed the best across the board. While the recall of the neural networks was the highest at 0.89, bearing the lowest F1 score (0.50) and lowest precision (0.35) makes the model unreliable. For consistent scores, the XGBoost performed the best, producing a recall of 0.70 and F1 score of 0.54. The XGBoost model performed the best because of its ability to capture the non-linear relationships involved with flight delays; there is not necessarily a linear relationship between temperature, precipitation, and dewpoint as illustrated with the misclassifications listed below. Say there is a day with low temperatures, high dewpoint, and low precipitation, and then those measurements flip, a logistic regression model would not be able to handle those types of complex, non-linear relationships. The ensemble of 150 decision trees also work in tandem to reduce the misclassification. If there are 50 decision trees that incorrectly classify a flight's outcome, there are still 100 other ones to compensate for the error.

In addition, the parameter tuning mentioned above illustrates the complexity of the XGBoost model. The sequential tuning of parameters like the learning rate, depth of trees, learning rate, and subsampling provides dynamic versatility that most other models lack. The logistic regression, while receiving lasso regression, the c parameter, and max iteration parameters, is still unable to match the versatility of the XGBoost model.



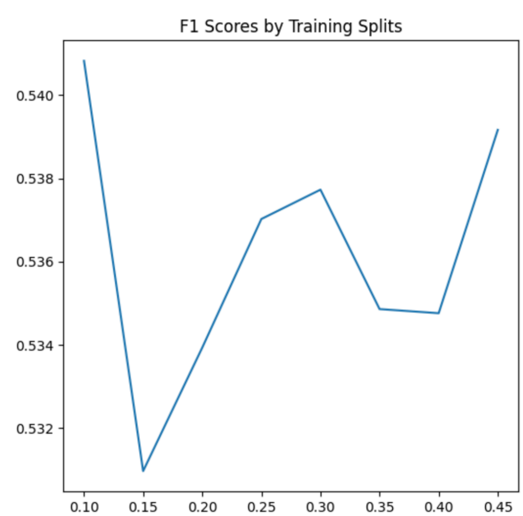
The largest decision in the model fitting made was lowering the decision threshold to 0.45. By just decreasing the threshold by 0.05, the XGBoost's model recall improved by almost 250%. While some may argue that this threshold causes the model to favor one decision over another, this decision was arrived at because of the nature of cancellations. When airlines have hundreds of lives in their hands in suboptimal conditions, they will tend to air on the side of caution when it comes to the decision of delaying takeoffs. Because airlines tend to favor one decision instead of another when bringing safety into consideration, the slight decrease in the decision threshold for all models is justified. This threshold proved to not be beneficial for the logistic regression model, as recall jumped to a perfect 1, and its precision jumped to 0. The MLP also saw an increase in recall, just not as drastic as the other two models.

While the MLP had the largest recall amongst all models, its accuracy was abysmal, and its precision was the lowest of the three models. Lowering the threshold proved beneficial in increasing recall, but that caused the accuracy and precision to drop. As mentioned previously, the decision to utilize the F1 score as the main metric for these models balances the performance of recall and precision to give a stronger understanding of a classification model's under-the-hood performance. The MLP produced the lowest F1 score of the three models. Neural networks are prone to settling on aggressive prediction strategies, and with the decision to limit the runtime of the model, the MLP learned quickly to predict delay more often than not, and that is what led to the high recall, but atrocious accuracy score.

Sensitivity analysis

To determine the model's underlying performance, and sensitivity, the imbalanced dataset with only 30% of the flights present in the set being delayed seemed like a viable option to explore. A subset of the data was created with an even 50/50 split of flights that were delayed and that took off on time. Nothing else was changed in the model other than the balanced and imbalanced data sets. When the model was run on this new subset, the accuracy dropped from 0.65 to 0.57, but the F1 score increased from 0.54 to 0.69, the precision increased from 0.44 to 0.55, and most impressively, the recall increased from 0.70 to 0.93. This indicates that when given balanced data, the model is able to correctly predict 93% of the flights that get delayed correctly. These increases in precision, recall, and F1 scores and the decrease in accuracy further validate the decision to not utilize accuracy as the main evaluation metric for the models. With the heavily imbalanced data, the models could more easily classify flights to take off on time and have a higher hit rate. When the model is trained on an even split, however, it is much more difficult to make this decision.

The finalized XGBoost model was also tested on the imbalanced dataset with varying sizes of training data splits. The test splits varied from 0.1 to 0.5. The highest F1 score of the model came from a test size of 0.1. This is not a viable split to use, though, as training the model on 90% of the data can lead to overfitting. The next highest F1 score came with a test size of 0.5. This also seemed too random in the decision making process for the models, as there was still half of the possible data that remained unseen. Therefore, the decision was made to use the test size of 0.3 to test the model. The little variability in F1 scores (as illustrated in the visual right) indicates that the model is moderately sensitive to shifts in the amount of data present for training, but is more robust in handling the different train/test split sizes.



This is encouraging to see these results, especially in the scope of this project, where only one month of data for one airport is being utilized. If the training data size grew, the split sizes can remain consistent with what splits were used for this analysis.

Failure Analysis

Record 282: Both the logistic regression model and XGBoost model misclassified this delayed flight as one that takes off on time. This is most likely because of the low dewpoint, lack of precipitation, and extremely cold temperatures. Even though there is no active precipitation at the time of the flight departure, the extremely cold temperatures can cause icing issues on runways, equipment failures, and other conditions that the model might not realize when determining delays. In addition, this flight is departing close to midnight, making the conditions at the end of the day in the dark more impactful as well.

Record 306: Another record that was misclassified consistently was the flight in row 306. Similar to the one in 282, this flight departed late at night, had an extremely low dewpoint, and had a low origins temperature. As this is the second record with these conditions to be misclassified, it is becoming more apparent that the models associated low temperatures and low dewpoint as traits of flights that take off on time. What can also cause the misclassification is the fact that the flights take off so close to midnight. This indicates that these flights are some of the last to depart from O'Hare, meaning that there is a smaller subset of flights leaving at this time, indicating that the model has not received sufficient training data for flights in this time slot. This makes sense, as the majority of flights transpire in the early morning throughout the early evening.

Record 315: Another record that the model misclassified was the one in row 315. This data point differs from the ones in 282 and 306 because this one occurs in the middle of the day. A potential reason for misclassification across all models was the fact that this record had an average temperature for the month of January, no precipitation, but an extremely low dew point. Due to the large volume of flights that occur in the afternoon that do not experience delays, the models learned these features to be consistent with flights that took off on time. The model fails to recognize here that the extremely low dewpoint can cause icing issues that the lack of precipitation does not identify. In a similar manner, ice can remain on the ground for hours to days in the cold if the temperature never goes above freezing, so if thawing does not occur, the ice will remain: something the models do not account for.

To help the model better identify these misclassifications in the future, further improvements could be made to the model by including another predictor with total precipitation from the last 24 hours. This could help the model learn, especially in January months, that residual ice can impact future flights just as much, if not more, than the precipitation that occurs at the time of departure. To combat the lack of data for late night flights, more data from surrounding months could help improve fitting for that time slot.

Unsupervised Learning

For the unsupervised learning portion of the project, we used clustering to identify hidden patterns in flight delays across airports, airlines, routes, and scheduling conditions. Our goal was not to predict a labeled outcome, but to group similar entities based on shared delay behavior, weather exposure, and operational characteristics. We began with the cleaned BTS and NOAA dataset and selected features related to delay outcomes, weather conditions, route structure, and departure timing. In addition to the original variables, we engineered several features to better capture disruption risk. These included origin and destination bad weather indicators, an overall any_bad_weather feature, morning and late-flight indicators, and a categorical time_of_day variable. We also created a weather delay sensitivity feature defined as the difference between delay rates during bad weather and normal weather. This was especially important because it allowed us to cluster airports or routes not only by how often delays occurred, but by how strongly the weather appeared to influence those delays.

To make the clustering more interpretable, we aggregated the flight data into several representations, including airport, airline, route, time of day, airport by time of day, airline by time of day, route airline time, and weather sensitivity route summaries. For each representation, we computed summary statistics such as delay rate, total flights, average distance, bad weather share, wind speed, precipitation, humidity, and the share of morning or late flights. These feature sets were chosen because they reflected the main drivers of delay discussed in our project of weather, congestion, route structure, and temporal scheduling effects. Before modeling, all numeric features were standardized using StandardScaler, and flight counts were log-transformed to reduce skew. Missing numeric values were imputed using median values.

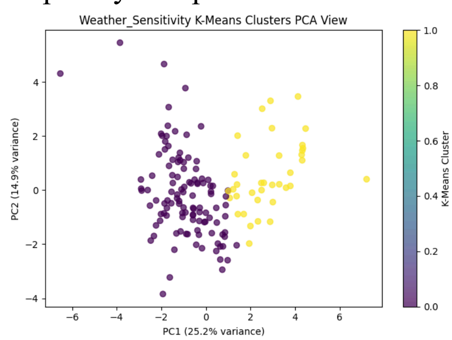
We explored two unsupervised methods, K-Means clustering and Hierarchical Agglomerative clustering with Ward linkage. We chose K-Means because it is a simple and widely used centroid-based clustering method that provides a strong baseline and supports clear clustering profiling. However, because K-means assumes relatively compact, spherical clusters, we also used hierarchical clustering as a second method with a very different underlying mechanism. Hierarchical clustering does not rely on centroids and is better able to capture irregular groupings, which we expected in flight delay data due to the interaction of weather, time, and operational factors. Using both methods allowed us to compare a partition based model with a hierarchy based model and better assess the structure of the data.

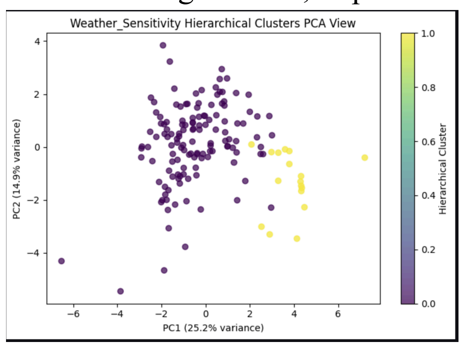
For both methods, we tuned the number of clusters (k) by tested values from 2 to 8 for each aggregated dataset, subject to the size of the representation. We selected the best configuration using the silhouette score, which measures how well observations fit within their assigned clusters compared to other clusters. We chose silhouette score as our primary metric because unsupervised learning has no ground truth labels, and the metric provides a standard way to evaluate both cluster cohesion and separation. For K-Means, we also recorded inertia as a secondary diagnostic, though silhouette score was used for model selection because it is more directly interpretable and allows comparison across both clustering families.

Overall, the results showed that clustering quality depended heavily on the feature representation. The strongest K-Means result came from the time_of_day representation, which achieved a silhouette score of 0.377 with 2 clusters. The strongest hierarchical result came from the airport_time representation, which achieved a silhouette score of 0.742 with 2 clusters. This was the best result across all experiments and suggests that airport behavior varies strongly by time of day. More moderate but still interpretable results appeared in the route and weather_sensitivity representations, where hierarchical clustering outperformed K-Means. In contrast, airline level and route airline time clustering produced weaker scores, suggesting that those representations were either too broad or too granular to reveal strong natural structure.

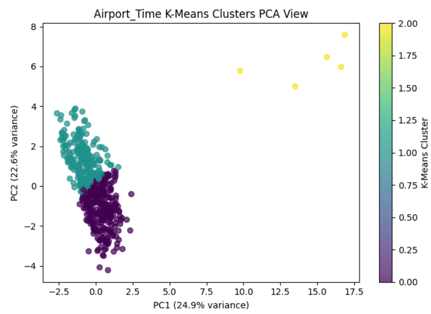
name	rows	n_features	best_kmeans_k	best_kmeans_silhouette	best_hierarchical_k	best_hierarchical_silhouette	_deepnote_index_column
time_of_day	6	14	2	0.377376382	2	0.377376382	3
route	151	17	2	0.209243333	2	0.278507945	2
airline	14	16	2	0.209046226	2	0.22173856	1
weather_sensitivity	151	18	2	0.204733162	2	0.28878326	7
airline_time	63	15	5	0.195606729	5	0.194071811	5
airport	151	18	2	0.192591729	3	0.17548774	0
airport_time	469	14	3	0.189009491	2	0.741603339	4
route_airline_time	766	14	2	0.156855215	2	0.120306965	6

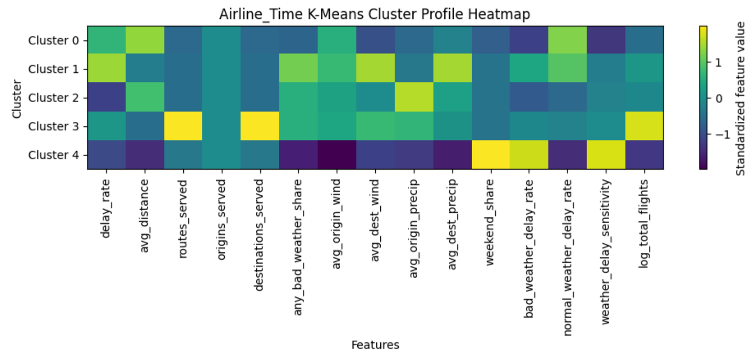
The weather sensitivity route clusters were especially meaningful in interpretation. K-Means separated routes into one cluster with higher delay rates, greeted bad weather exposure, strong weather delay sensitivity, and more late flights, and another cluster with lower delay rates, more morning flights, and much weaker weather sensitivity. Although the silhouette score for this representation was only moderate, the clusters were still operationally useful because they reflected realistic differences in disruption risk. More generally, hierarchical clustering often matched or outperformed K-Means, indicating that the underlying structure of the data was not always well represented by spherical clusters. This supports the idea that delay behavior is shaped by complex and uneven interactions among weather, airport activity, and scheduling.





We also performed a sensitivity analysis by comparing clustering quality across different values of (k) and across different feature representations. In most cases, the best results occurred with only 2 or 3 clusters, suggesting that the data contained a few broad operational groupings rather than the many sharply distinct subgroups. The results were also clearly sensitive to how the entries were represented. For example, airport_time performed much better than airport-only clustering, showing that adding temporal context significantly improved separation. Similarly, weather-focused route representations produced more meaningful clusters than route structure alone. To summarize these results visually, we used silhouette score plots across values of (k), cluster profile heatmaps, and PCA-based clustering visualizations. Together, these results show that unsupervised learning was most effective when the feature representation captured time of day and weather related variation, and that hierarchical clustering provided the strongest overall grouping structure in our analysis.





Discussion

What did you learn from doing Part A? (4 points)

In the supervised learning part of this project, it was surprising to find such difficulty in achieving high evaluation metrics. The initial evaluation of the models and parameter tuning were conducted with accuracy, which was somewhat simple to get close to 80%. However, upon looking at the classification report, it quickly became apparent that the model was performing quite poorly when it came to identifying the positive cases of flights that experienced a delay. As accuracy improved, the models' recall scores all ranged below 10%. Pivoting to the F1 score helped improve the overall performance of the model.

Similar to what was surprising, the biggest challenge encountered was the imbalanced data set that we worked with. Almost 70% of the flights that departed from the O'Hare airport in January did not experience a delay. Because of the amount of data that favored flights not being delayed, all three supervised learning models adapted to predict "no delay" if there was any uncertainty. The accuracy was able to improve, but the main objective of the models was being mitigated. The first way to respond to the imbalanced data was to work on optimizing the recall of the models. By optimizing recall, accuracy had to be heavily sacrificed. For example, the logistic regression model with tuned parameters could achieve as high as 0.78 accuracy, but the recall was a mere 0.09. As recall increased across the three models, the accuracy dropped. Despite using F1, precision, and recall to evaluate the model performances, it was still tough to achieve high F1 scores. The XGBoost model obtained the highest F1 of 0.54. The XGBoost and MLP models combatted the challenge with the response of the lower decision threshold of 0.45. This decision outlined above to "air on the side of caution" like airlines do when considering delays helped get the F1 scores to the ranges seen in the summary table of findings. This response allowed for more delayed flights to be predicted correctly. The MLP model was able to predict 89% of delayed flights correctly.

Even though the models were able to better optimize the prediction of delayed flights, the overall model performances were slightly above average. The small amount of delayed flights proved to be detrimental in overall model performance. If there was more time for this project, the solution could be improved with a larger amount of data, both in flights departing from O'Hare in other months, but also from other major airports. Nearly 75% of flights from all American airports do not experience a delay of more than 15 minutes, though (Latham, *Why flights are more likely to be delayed at certain times of day*). The imbalanced data on flight delays will continue to be skewed as more months and airlines are added, but the sheer volume of flights should help smooth the model performance out better. Our objective was to create supervised learning models that predicted flight delays correctly. The bright spot of the hypothesis was that a model could predict 89% of the delayed flights correctly, but the low accuracy and F1 scores illustrate room for model improvement in future iterations on this project.

In regards to the Unsupervised Learning results, it was surprising to see that the more detailed clustering groups did not provide the more accurate scores. We expected to see more meaningful clusters from the route + airline+ time of day model since it provided the most detail, however we saw that the silhouette score was much lower than some of the broader models, making it seem that sometimes having more data does not always mean more accurate results. After further inspection, it was revealed that the route + airline+ time of day model combinations become too specific, providing fewer available flights for the model to utilize.

One challenge we faced was interpreting the models silhouette scores. While we noticed that many provided vast differences in the silhouette scores, around .15 and .29, meaning the clusters were not perfectly separated. Thus, we interpreted this variability as the fact that delays are influenced by many factors such as weather, airport traffic, airline operations, route distance, and time of day. To better understand these scores and the variability, we used clutter profile heatmaps and PCA scatterplots, which helped demonstrate which features were above or below average for each cluster, making it easier to describe clusters as high delta, weather sensitive, or lower risk groups.

If we had more time and resources, we would definitely love to extend our dataset to include a longer timeframe for our dataset, which would improve the cluster reliability and reduce the risk that our current sample provides. We would also love to add more complex weather features to index more features such as wind speed and precipitation, or even operational features such as airport congestion, gate availability, and air traffic control information, which would explain delay metrics more accurately.

Ethical Considerations

Our supervised learning section raises ethical concerns related to model reliability, bias, and interpretation. The supervised model predicts flight delays and cancellations from historical data, and because of this, it may perform better for some airports, airlines, routes, or weather conditions than others. This creates a risk of uneven performance and historical bias, where the model learns patterns already present in past operations and reinforces them. There is also a risk that users may treat predictions as certain rather than probabilistic, even though the model only identifies patterns associated with delays and does not establish causation. These issues could be addressed by clearly communicating model uncertainty, reporting performance across different subgroups, and emphasizing that the results are decision support tools rather than guarantees or explanations of why a delay occurs.

The supervised learning model also includes ethical concerns of misuse of the output. If predictions are used to guide traveler behavior or suggest whether to purchase flight insurance, inaccurate or poorly calibrated predictions could influence decisions in misleading ways. Users may also draw conclusions the model was not designed to support, such as assuming a specific airline or airport is inherently bad based only on model outputs. These concerns can be addressed by presenting predictions with probability estimates and limitations, avoiding overly deterministic language, and including clear disclaimers that the model should not be the sole basis for financial or travel decisions.

The unsupervised model raises ethical concerns mainly related to interpretation and downstream impact of the results. Clustering airports, airlines, or routes may produce groupings that appear meaningful, but those patterns do not necessarily explain causes of delays. There is a risk of confusing correlation with causation or oversimplifying operational behavior by assigning broad labels to groups of airports or airlines. If these clusters were shared publicly or used in decision making, they could influence traveler behavior or unfairly affect airline or airport reputation and revenue.

Another concern in the unsupervised model is accountability. If clustering results are used to support planning or policy decisions, it may be unclear who is responsible when those conclusions are wrong or misleading. Since unsupervised methods are exploratory, the results should be framed carefully as descriptive patterns rather than firm conclusions. These issues can be addressed by using cautious language, avoiding stigmatizing labels for clusters, and clearly stating that the results are exploratory summaries meant to support analysis rather than direct operational or financial decisions.

Statement of Work

Each member of our group played an important role in the project. Gabrielle was responsible for data collection and cleaning, Brendan developed the supervised learning model, and Siddarth developed the unsupervised learning model. All three team members also contributed to writing and preparing the final report.

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